Tire Management System

Functional & System Design

Administrasi Data Master

- **Vehicle**
- Vehicle Trail Setup
- Tire

Admin Master Vehicle

CRUD (Create, Read, Update, Delete) data master vehicle dan segmen/bagian dari trailer

- Jenis:
  - Car (4 roda) …… [CAR]
  - Light Truck (4 roda) …… [LTRUCK]
  - (Double) Truck (6 roda) ……[TRUCK]
  - Trailer-Head (6-10 roda) …… [THEAD]
  - Trailer-Chassis (12-16 roda)…[TCHASS]
  - Trailer-Dolly (8-12 roda) ……[TDOLLY]
- Proses: create, read (view list/table & view details), update, delete

Admin Master Vehicle

[Page] View List/Table

All | Car | Truck | Trailer Head | Trailer Chassiss | Trailer Dolly

No	ID	Brand	Model (Year)	Plat No	Frame SN	Nb Wheels
1	TCHAS-001	Hino	407 GT (2015)	-	XDC123454	16
2	TCHAS-002	Kobelco	Heavy Duty XT (2011)	-	TRB131359	12
3	THEAD-001	Scania	Avenger T-1 (2022)	B 12345 FN	SCN432567	10
4	TDOLL-001	Kobelco	KB-12345 (2016)	-	TRB847365	12
5	LTRUCK-001	Volvo	Cyber Truck (2022)	B 6789 XA	VOL432567	4
6	CAR-001	Toyota	Hilux GTR	B 674538 TYE	TRB847365	4

Admin Master Vehicle

[Page] View List/Table

All | Car | Truck | Trailer Head | Trailer Chassiss | Trailer Dolly

THEAD-001

Scania SC-1 (2019)

25.783 Km

B 123456 FN | 10 Wheeler

Frame No
SC1234324

TDOLLY-001

Kobelco Falcon Heavy (2015)

15.744 Km

- | 12 Wheeler

Frame No
KB1234324

LTRUCK-002

Volvo Cyber Truck (2019)

6.890 Km

B 123456 FN | 10 Wheeler

Frame No
VL1234324

Admin Master Vehicle

[Page] View Details

THEAD-001

Manage Tire

Status

Kind
Trailer Head

25.783 Km

Brand & Model
Scania SC-1 (2019)

Number of Wheels
10

Plate No
B 54357 FV

Frame Serial Number
FG467312644932

Tire Data

Mobility Data

Edit

Delete

Manage tire? *Untuk kasus Bayan Resource unit Trailer Headnya tidak ditracking dan tidak dimanage bannya karena ukurannya berbeda dengan ukuran ban Chassis dan Dolly*

Dengan demikian, data tentang Trailer Headnya ***boleh jadi tidak dientry*** *ke dalam sistem*

Admin Master Vehicle

[Page] Create & Update

Click to upload new image

THEAD-001

Manage Tires

Status

Kind

Trailer Head

Brand & Model

Scania | SC-1 | 2019

Number of Wheels

- | 10 | +

Plate No

B 544357 FV

Frame Serial No

FGX123490432

Tracker Device ID

FGX123490432

Save

Delete

Alt A

Dengan layout persis View Details

Alt B

THEAD-1

Status

Kind

Please select

Car
Light Truck
(Double) Truck
Trailer-Head
Trailer-Chassis
Trailer-Dolly

Brand

Scania

Model/Year

HD 203 (2019)

Number of Wheel

- | 10 | +

Manage Tires

Plate No

B 86750 FV

Frame Serial Number (Nomor Rangka)

475F89BXP

Upload File | No file chosen

Save

Delete

Admin Master Vehicle

Data Definition & Data Sample

- **Kind ..... Prefix ID**
  - Car (4 roda) ..... [CAR]
  - Light Truck (4 roda) ..... [LTRUCK]
  - (Double) Truck (6 roda) .....[TRUCK]
  - Trailer-Head (10 roda) ..... [THEAD]
  - Trailer-Chassis (12-16 roda)...[TCHASS]
  - Trailer-Dolly (8-12 roda) .....[TDOLLY]
- **Brand/Make**
  Merk Kendaraan
- **Model**
  Model Kendaraan
- **Year**
  Tahun model kendaraan
- **Plate No**
  Nomor Plat Polisi
- **Frame Serial No**
  Nomor rangka
- **Number of Wheels**
  Jumlah roda

ID	Kind	Brand/Make	Model	Year	Plate No	Frame No	Nb Wheel
TCHASS-001	Trailer Chassis	Kobelco	Falcon Heavy 5	2022	-	2748249284u	16
LTRUCK-004	Light Truck	Scania	Cyber Truck X	2023	B 34048 BVF	487239847328	4

Administrasi Data Master

- Vehicle
- **Vehicle Trail Setup**
- Tire

Admin Master Vehicle Trail Setup/Configuration

CRUD (Create, Read, Update, Delete) rangkaian kendaraan gandeng

- Tipe
  - Single Trailer [VTS]
  - Double Trailer [VTD]
- Trailer Head [THEAD] as Head
- Proses: create, read (view list/table & view details), update, delete

Admin Master Vehicle Trail Setup

[Page] View List/Table

All | Non Trailer | Trailer | Double Trailler

No	ID VTSeteup	Head	Brand & Model	Plate No	Trailers	Total Wheels
1	VTD-001	THEAD-002	Scania Heay Duty (2022)	B 12345 FN	TCHASS-003, TDOLLY-005, TCHASS-007	44
2	VTS-001	THEAD-001	Scania Avenger T-1 (2022)	B 12345 FN	TCHASS-010	22
3	VTD-001	THEAD-005	Scania Heay Duty (2022)	B 12345 FN	TCHASS-013, TDOLLY-021, TCHASS-009	44
4	VTS-002	THEAD-007	Scania Avenger T-1 (2022)	B 12345 FN	TCHASS-011	22
5	VTS-003	THEAD-004	Volvo Cyber Truck (2022)	B 6789 XA	TCHASS-021	22
6	VTS-004	THEAD-006	Scania Avenger T-1 (2022)	B 12345 FN	TCHASS-017	22

Admin Master Vehicle Trail Setup

[Page] View List/Table

All | Trailer | Double Trailler

Single Trailer
VTS-001

THEAD-001
Scania SC-1 (2019)
B 123423 FN,
10 Wheeler

TCHASS-003
Kobelco Falcon Heavy (2015)
12 Wheeler

22
Wheels

Double Trailer
VTD-001

THEAD-002
Scania SC-1 (2019)
B 123423 FN,
10 Wheeler

TCHASS-003
Kobelco Falcon Heavy (2015)
16 Wheeler

TDOLLY-005
Kobelco Falcon Heavy (2015)
12 Wheeler

TCHASS-007
Kobelco Falcon Heavy (2015)
16 Wheeler

44
Wheels

Single Trailer
VTS-002

THEAD-003
Scania SC-1 (2019)
B 123423 FN,
10 Wheeler

TCHASS-002
Kobelco Falcon Heavy (2015)
12 Wheeler

22
Wheels

Admin Master Vehicle Trail Setup

Double Trailer

VTD-001

44
Wheels

THEAD-001
Scania SC-1 (2019)
B 123423 FN,
10 Wheeler

Tire Data | Mobility Log

TCHASS-003
Kobelco Falcon Heavy
(2015)
16 Wheeler

Tire Data | Mobility Log

TDOLLY-005
Kobelco Falcon Heavy
(2015)
12 Wheeler

Tire Data | Mobility Log

TCHASS-002
Kobelco Falcon Heavy
(2015)
16 Wheeler

Tire Data | Mobility Log

Edit | Delete

[Page] View Details

Admin Master Vehicle Trail Setup

Double Trailer

VTD-001

Tipe

Double Trailer

44
Wheels

Head

THEAD-001 Scania SC-1 (2019), B 123423 FN, 10 Wheeler

HEAD-1
Scania SC-1 (2019), B 123423 FN, 10 Wheeler

THEAD-2
Scania SC-1 (2019), B 123423 FN, 6 Wheeler

Trailer

TCHASS-001 Kobelco SC-1 (2019) 16 Wheeler

TDOLLY-003 Hino XYZ2 (2019), 12 Wheeler

+ Trailer

Save Delete

[Page] Create & Update

Administrasi Data Master

- Vehicle
- Vehicle Trail Setup
- **Tire**

Admin Master Tire

CRUD (Create, Read, Update, Delete) data master ban

- Data:
  - ID: X-{YY pembelian} {3 digit urutan ban di tahun YY}
  - No. Seri
  - Brand/Make
  - Size
  - Current Status: New, Attached, Detached, In Repair, Retired
  - Provisioning Date (tanggal pembelian)
- Proses: create, read (view list/table & view details), update, delete

X-24019

Tahun 2024 no urut **19**

Admin Master Tire

[Page] View List/Table

All | New | Attached | Detached | In Repair | Retired

No	ID	Brand	Status	Mileage (KM)	Size	Serial Number	Provisioning Date
1	X-22104	Toyo HeavyDuty	New	0	635/60/R30	XDC123454	2022-03-15
2	X-19135	Toyo HeavyDuty	Attached to TCHASS-03	34.390	635/60/R30	TRB131359	2022-03-15
3	X-21001	Toyo HeavyDuty	In Repair	8.945	635/60/R30	SCN432567	2022-03-15
4	X-20015	Dunlop Monster	Attached to TCHASS-03	97.213	635/60/R30	TRB847365	2022-03-15
5	X-18101	Michelin Monster	Retired	135.176	635/60/R30	VOL432567	2022-03-15
6	X-17097	Bridgestone Turanza	Detached	4.218	635/60/R30	TRB847365	2022-03-15

Admin Master Tire

[Page] View List/Table

All | New | Attached | Detached | In Repair | Retired

X-23001
Toyo HeavyDuty
Size: 645/50/R35
Provisioning Date: 22 Agustus 2023
0 Km
Status: New
Serial No
SC1234324

X-19035
Bridgestone Turanza
Size: 645/50/R35
Provisioning Date: 22 Agustus 2019
25.783 Km
Status: Attached to TCHASS-01
Serial No
XL482903

X-20016
Michelin Monster
Size: 645/50/R35
Provisioning Date: 22 Agustus 2020
13.188 Km
Status: Attached to TCHASS-01
Serial No
TG8129034

Admin Master Tire

[Page] View Details

X-23001

Toyo HeavyDuty

Mileage
17.893 Km

Serial No
SC1234324

Size
645/50/R35

Provisioning Date
22 Agustus 2023

Status
Attached to TCHASS-015

Tire Log & Mileage Details

Edit

Delete

Admin Master Tire

[Page] Create & Update

X-23001

Toyo HeavyDuty

Mileage
17.893 Km

Serial No
SC123424

Size
645/50/R35

Provisioning Date
22 Agustus 2024

Status
Attached

TCHASS-1

Tire Log & Mileage Details

Save

Delete

Day to Day Operations

- **Dashboard**
- Vehicle/Truck & Trailer
- Tire

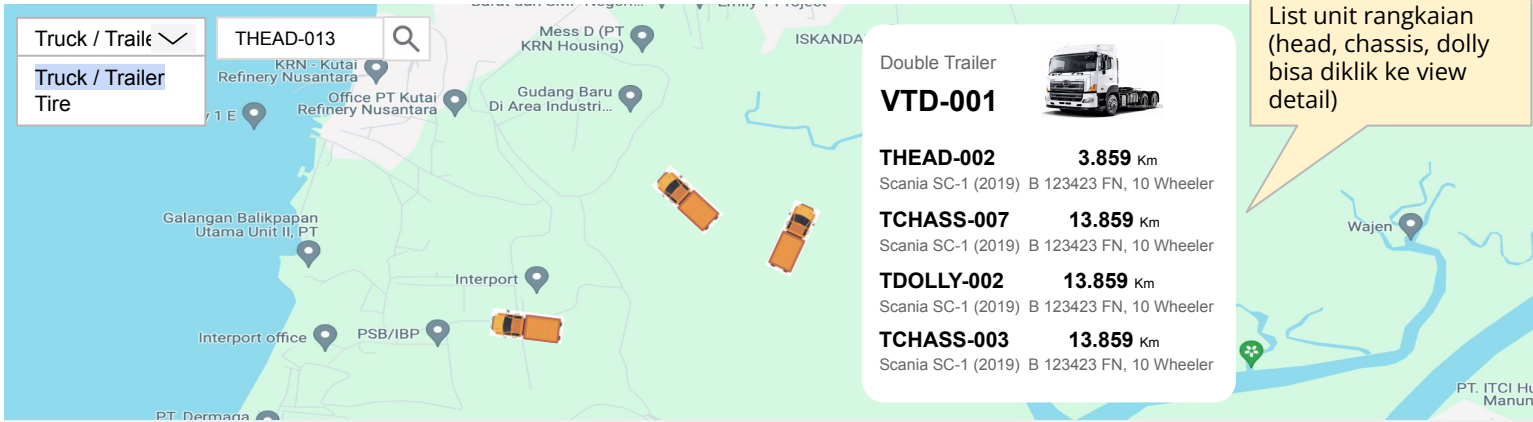
Dashboard

Map (with search function) & basic statistic of assets and their status

- **Dashboard**
- Truck & Trailer
  - Trailed
  - Units
- Tire
- Administration
  - Truck & Trailer
  - Truck Setup
  - Tire

{ or more generic menu}

- **Dashboard**
- Vehicle
  - Trailed
  - Units
- Tire
- Administration
  - Vehicle
  - Vehicle Setup
  - Tire



Popup on click pada icon truck

List unit rangkaian (head, chassis, dolly bisa diklik ke view detail)

Number of Truck Head

218

- 192 Active
- 16 Inactive

Number of Truck Chassis

512

- 192 Linked
- 16 INot linked

Number of Truck Dolly

197

- 192 Linked
- 16 INot linked

Number of Tires

7643

- 192 New
- 16 Attached
- 32 Detached
- 65 Ub Repair

Today's Top 5 Truck Mileage

THEAD-023	127 Km
THEAD-142	108 Km
THEAD-041	97 Km
THEAD-002	96 Km
THEAD-201	87 Km

Day to Day Operations

- Dashboard
- **Vehicle/Truck & Trailer**
  - **Trailed**
  - All Unit
- Tire

Truck & Trailer » Trailed

[Page] View List/Table
Similar to Vehicle Page for Administrator

All | Trailer | Double Trailler

Single Trailer
VTS-001

THEAD-001
Scania SC-1 (2019)
B 123423 FN,
10 Wheeler

TCHASS-003
Kobelco Falcon Heavy (2015)
12 Wheeler

22
Wheels

Double Trailer
VTD-001

THEAD-002
Scania SC-1 (2019)
B 123423 FN,
10 Wheeler

TCHASS-003
Kobelco Falcon Heavy (2015)
16 Wheeler

TDOLLY-005
Kobelco Falcon Heavy (2015)
12 Wheeler

TCHASS-007
Kobelco Falcon Heavy (2015)
16 Wheeler

44
Wheels

Single Trailer
VTS-002

THEAD-003
Scania SC-1 (2019)
B 123423 FN,
10 Wheeler

TCHASS-002
Kobelco Falcon Heavy (2015)
12 Wheeler

22
Wheels

Truck & Trailer » Trailed

Double Trailer

VTD-1

44
Wheels

THEAD-1
Scania SC-1 (2019)
B 123423 FN,
10 Wheeler

Tire Data | Mobility Log

TCHASS-3
Kobelco Falcon Heavy
(2015)
16 Wheeler

Tire Data | Mobility Log

TDOLLY-5
Kobelco Falcon Heavy
(2015)
12 Wheeler

Tire Data | Mobility Log

TCHASS-3
Kobelco Falcon Heavy
(2015)
16 Wheeler

Tire Data | Mobility Log

[Page] View Details Truck & Trailer
Similar to Vehicle Page for Administrator

When clicked go to this unit detail view

Day to Day Operations

- Dashboard
- **Vehicle/Truck & Trailer**
  - Trailed
  - **Units**
- Tire

Truck & Trailer » Units

[Page] View List/Table
Similar to Vehicle Page for Administrator

All | Car | Truck | Trailer Head | Trailer Chassiss | Trailer Dolly

THEAD-001
Scania SC-1 (2019)

25.783 Km
B 123456 FN | 10 Wheeler

Frame No
SC1234324

TDOLLY-001
Kobelco Falcon Heavy (2015)

15.744 Km
- | 12 Wheeler

Frame No
KB1234324

LTRUCK-002
Volvo Cyber Truck (2019)

6.890 Km
B 123456 FN | 10 Wheeler

Frame No
VL1234324

Truck & Trailer » Units

[Page] View Details

THEAD-001

25.783 Km

Status: Active

Manage Tire: On

Kind
Trailer Head

Brand & Model
Scania SC-1 (2019)

Number of Wheels
10

Plate No
B 54357 FV

Frame Serial Number
FG467312644932

Trail | Mobility Log | Tire

Truck & Trailer » Units

[Tab] Trail

Trail	Mobility Log	Tire

Current Trail Setup

VTD-089

as Trailer Head

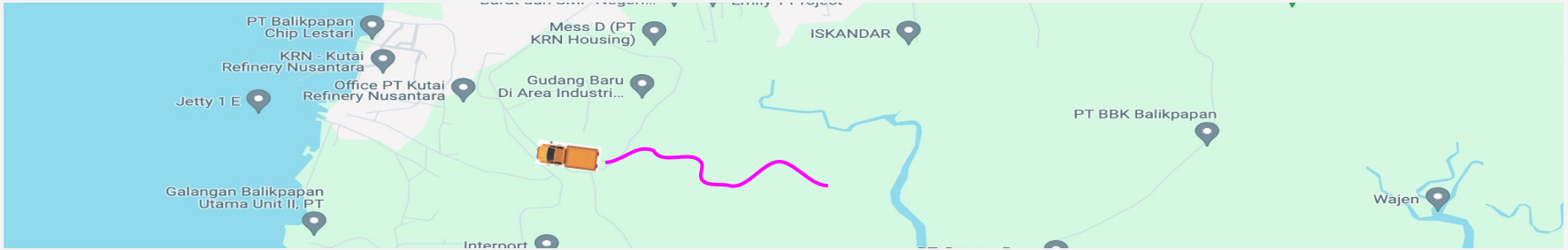
THEAD-1	Scania SC-1 (2019) B 123423 FN, 10 Wheeler
TCHASS-3	Kobelco Falcon Heavy (2015) 16 Wheeler
TDOLLY-5	Kobelco Falcon Heavy (2015) 12 Wheeler
TCHASS-7	Kobelco Falcon Heavy (2015) 16 Wheeler

When clicked go to this unit detail view

Truck & Trailer » Units

[Tab] Mobility Log

Trail | Mobility Log | Tire



Today | This Week | Last 30 Days

No	Datetime	Long	Lat	Speed (km/h)	Distance (m)
1	2024-07-15 12:30:00	23.432234432	23.432234432	10	25
2	2024-07-15 12:30:05	23.432234432	23.432234432	13	27
3	2024-07-15 12:30:10	23.432234432	23.432234432	15	30

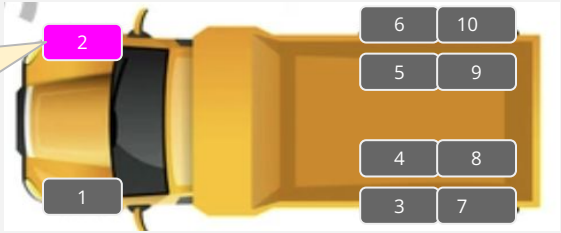
Truck & Trailer » Units

[Tab] Tire Data

Trail	Mobility Log	Tire

Onclick on picture selaras highlightnya dengan highlight di row

Ondoubleclik muncul pop up ganti ban



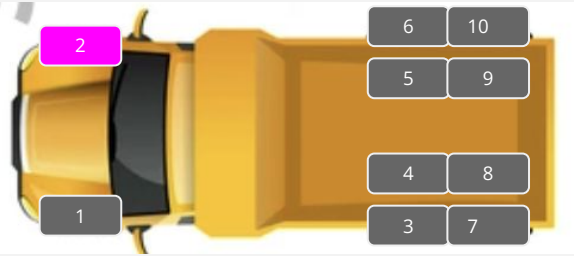
When clicked go to this unit detail view

No	Tire ID	Brand	Size	Mileage	Action
1	X-20011	Toyo Heavy Duty	645/50/R40	19.587	Change Tire
2	X-23089	Michelin Monster	645/50/R40	13.721	Change Tire
3	X-17022	Bridgestone Turanza	645/50/R40	150.5436	Change Tire

Truck & Trailer » Units

[Tab] Mobility Log

Trail | Mobility Log | Tire



No	Tire ID	Brand	Size		
1	X-20011	Toyo Heavy Duty	645/50/R40		
2	X-23089	Michelin Monster	645/50/R40	[illegible]	[illegible]
3	X-17022	Bridgestone Turanza	645/50/R40	150.5436	30

Change Tire

Tire Position
2

Date & Time
22 Agustus 2024 13:59

Current Tire
X-19018

New Tire
X-22109

Save | Cancel

Day to Day Operations

- Dashboard
- Vehicle/Truck & Trailer
  - Trailed
  - Units
- **Tire**

Tire

[Page] View List/Table

All | New | Attached | Detached | In Repair | Retired

X-23001
Toyo HeavyDuty
Size: 645/50/R35
Provisioning Date: 22 Agustus 2023
0 Km
Status: New
Serial No
SC1234324

X-19035
Bridgestone Turanza
Size: 645/50/R35
Provisioning Date: 22 Agustus 2019
25.783 Km
Status: Attached to TDOLLY-013
Serial No
XL482903

X-20016
Michelin Monster
Size: 645/50/R35
Provisioning Date: 22 Agustus 2020
13.188 Km
Status: Attached to TCHASS-071
Serial No
TG8129034

Tire

[Page] View Details

X-23001

Toyo HeavyDuty

Mileage
17.000 Km

Serial No
SC1234324

Size
645/50/R35

Provisioning Date
22 Agustus 2023

Status
Attached to TCHASS-095

When clicked go to this unit detail view

Tire Log | Mileage Details

Tire

[Page] View Details

Tire Log | Mileage Details

All | New | Attached to | Detached from | In Repair

No	Datetime	Activity	Unit ID	Unit Mileage
1	2024-07-01 09:00:00	Attached to	TCHASS-092	5.000
2	2024-06-15 16:47:00	In Repair	-	-
3	2024-06-15 06:47:00	Detached from	TCHASS-037	20.000
4	2022-03-07 06:47:00	Attached to	TCHASS-037	5.000
5	2022-01-12 16:47:00	New	-	-

Tire

[Page] View Details

Tire Log | Mileage Details

Misalnya per hari ini (17 Juli 2024) odo TCHASS-092 sudah 7.000

Sementara di 1 Juli masih di 5000 (lihat Tire Log di halaman sebeliumnya)

No	Unit ID	From	To	Mileage
1	TCHASS-092	2024-07-21 09:00:00	-	2.000
2	TCHASS-037	2022-03-07 06:47:00	2024-06-15 06:47:00	15.000

AIS Monitoring Dashboard

Functional & System Design

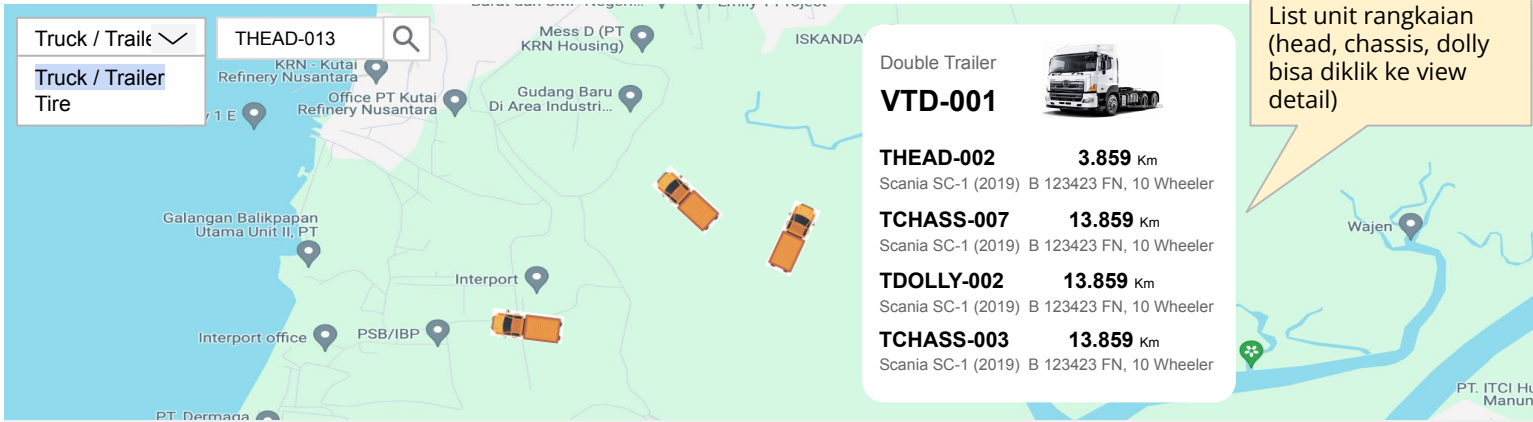
Home AIS

Dashboard

Map (with search function) & basic statistic of assets and their status

Popup on click pada icon truck

List unit rangkaian (head, chassis, dolly bisa diklik ke view detail)



Number of Truck Head

218

- 192 Active
- 16 Inactive

Number of Truck Chassis

512

- 192 Linked
- 16 INot linked

Number of Truck Dolly

197

- 192 Linked
- 16 INot linked

Number of Tires

7643

- 192 New
- 16 Attached
- 32 Detached
- 65 Ub Repair

Today's Top 5 Truck Mileage

THEAD-023	127 Km
THEAD-142	108 Km
THEAD-041	97 Km
THEAD-002	96 Km
THEAD-201	87 Km

Vessel Info >> Pop Up

[Page] View Details

MV. BUANA INDAH (31319845738)

Mon, 25 Nov 2024
12:27:01 GMT+07

Vessel ID
67DSYBDS7

Device ID
67DSYBDS7

Vessel Type
71 Cargo, Hazardous Cat. A

Longitude
119° 48’ 50.34” E

Latitude
119° 48’ 50.34” E

Speed
10

Heading
350

View More

Vessel Info >> Tab Vessel Info

[Page] View Details

MV. BUANA INDAH (31319845738)

Mon, 25 Nov 2024
12:27:01 GMT+07

Vessel ID
67DSYBDS7

Device ID
VDN73084420

Vessel Type
71 Cargo, Hazardous Cat. A

Longitude
119° 48’ 50.34” E

Latitude
119° 48’ 50.34” E

Speed
10

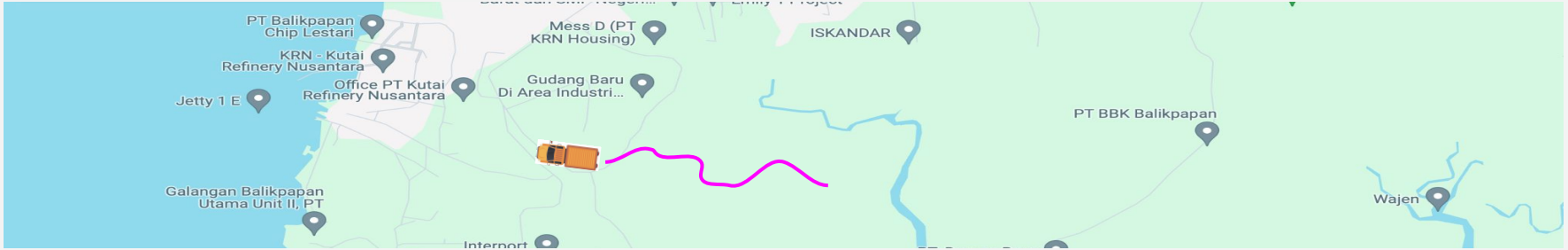
Heading
350

Positions | Geofencing Log

Truck & Trailer » Units

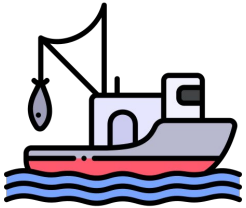
[Tab] Positions

Positions | Geofencing Logs

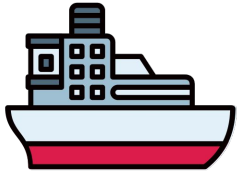


Today | This Week | Last 30 Days

No	Datetime	Long	Lat	Speed (km/h)	Distance (m)
1	2024-07-15 12:30:00	23.432234432	23.432234432	10	25
2	2024-07-15 12:30:05	23.432234432	23.432234432	13	27
3	2024-07-15 12:30:10	23.432234432	23.432234432	15	30



Fishing boat icon



Passanger boat